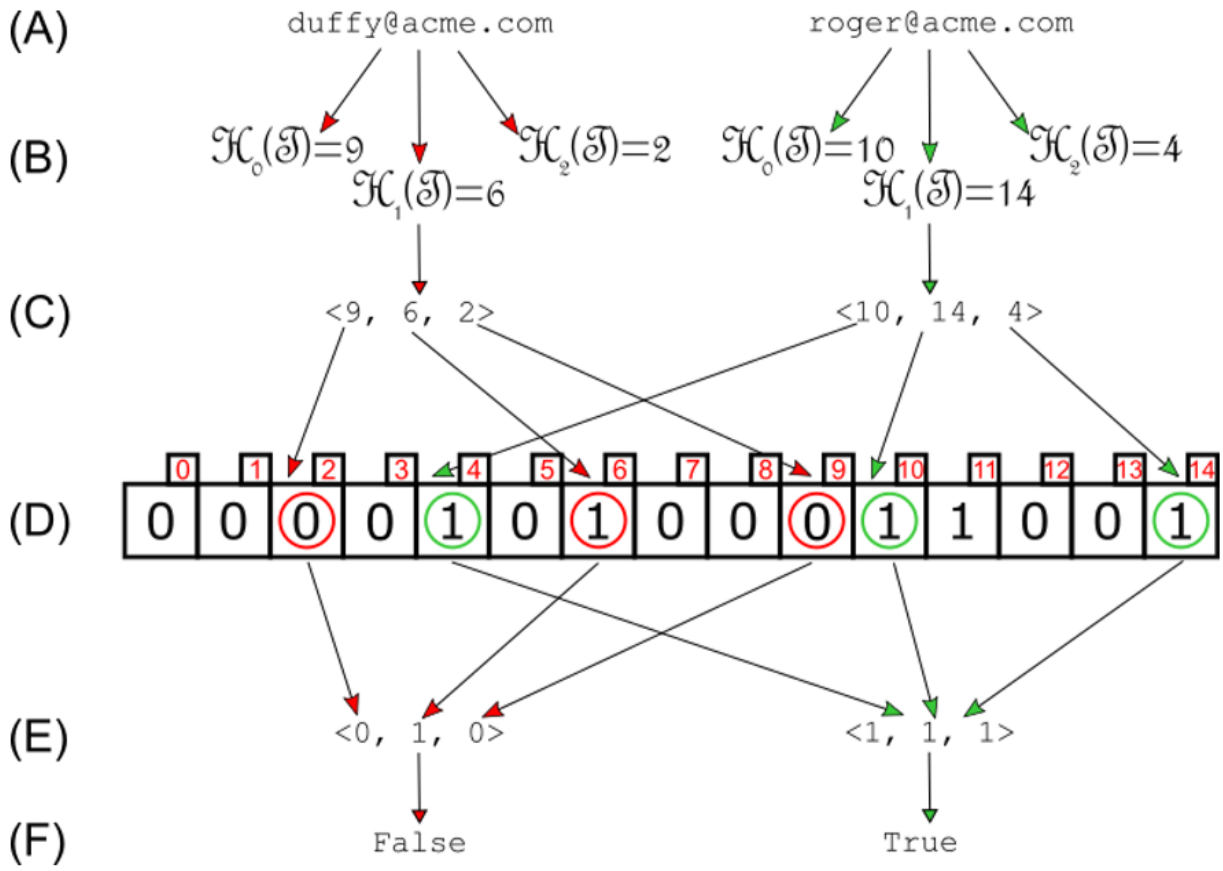


Figure 4.4 Checking if an entry is in a Bloom filter, step by step.
(A) We start with the email we would like to check, "`duffy@acme.com`".
(B) The key (our email) is processed through our set of hash functions: in our example we assume `k=3`, so we'll have 3 different hash functions, H_0, H_1 and H_2.
(C) Each hash function produces an index for our binary array: in this case the three indices might be, for instance, `<9, 6, 2>`.
(D) We access the elements of the binary array at those indices.
(E) The first element, at index `2`, is `0`. The other bits are `0` and `1` respectively.